```
import torch
import torch.nn as nn
import torch.optim as optim
import torchvision
import torchvision.transforms as transforms
from torch.utils.data import DataLoader
import matplotlib.pyplot as plt
import numpy as np
from sklearn.metrics import confusion_matrix, classification_report
import seaborn as sns
```

```
transform = transforms.Compose([
    transforms.ToTensor(),
    transforms.Normalize((0.5,), (0.5,))
])
```

```
train_dataset = torchvision.datasets.FashionMNIST(root="./data", train=True, transform=transform,
test_dataset = torchvision.datasets.FashionMNIST(root="./data", train=False, transform=transform,
```

```
100%|██████████| 26.4M/26.4M [00:01<00:00, 13.8MB/s]
100%|██████████| 29.5k/29.5k [00:00<00:00, 209kB/s]
100%|██████████| 4.42M/4.42M [00:01<00:00, 3.86MB/s]
100%|██████████| 5.15k/5.15k [00:00<00:00, 12.0MB/s]
```

```
image, label = train_dataset[0]
print(image.shape)
print(len(train_dataset))
```

```
torch.Size([1, 28, 28])
60000
```

```
image, label = test_dataset[0]
print(image.shape)
print(len(test_dataset))
```

```
torch.Size([1, 28, 28])
10000
```

```
train_loader = DataLoader(train_dataset, batch_size=32, shuffle=True)
test_loader = DataLoader(test_dataset, batch_size=32, shuffle=False)
```

```
class CNNClassifier(nn.Module):
    def __init__(self):
        super(CNNClassifier, self).__init__()

        self.conv1 = nn.Conv2d(1, 32, kernel_size=3, padding=1)
        self.conv2 = nn.Conv2d(32, 64, kernel_size=3, padding=1)
```

```
        self.pool = nn.MaxPool2d(kernel_size=2, stride=2)
        self.relu = nn.ReLU()

        self.fc1 = nn.Linear(64 * 7 * 7, 128)
        self.fc2 = nn.Linear(128, 10)

        self.dropout = nn.Dropout(0.25)

    def forward(self, x):

        x = self.pool(self.relu(self.conv1(x)))
        x = self.pool(self.relu(self.conv2(x)))

        x = x.view(x.size(0), -1)

        x = self.dropout(self.relu(self.fc1(x)))
        x = self.fc2(x)

        return x
```

```
!pip install torchsummary

from torchsummary import summary
import torch

model = CNNClassifier()

device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
model = model.to(device)

print('Name: kaaviyan.k')
print('Register Number: 212224240066')

summary(model, input_size=(1, 28, 28), device=str(device))
```

```
Requirement already satisfied: torchsummary in /usr/local/lib/python3.12/dist-packages (1.5.1)
Name: kaaviyan.k
Register Number: 212224240066
----------------------------------------------------------------
        Layer (type)               Output Shape         Param #
================================================================
            Conv2d-1           [-1, 32, 28, 28]             320
              ReLU-2           [-1, 32, 28, 28]               0
         MaxPool2d-3           [-1, 32, 14, 14]               0
            Conv2d-4           [-1, 64, 14, 14]          18,496
              ReLU-5           [-1, 64, 14, 14]               0
         MaxPool2d-6             [-1, 64, 7, 7]               0
            Linear-7                  [-1, 128]         401,536
              ReLU-8                  [-1, 128]               0
           Dropout-9                  [-1, 128]               0
           Linear-10                   [-1, 10]           1,290
================================================================
Total params: 421,642
Trainable params: 421,642
Non-trainable params: 0
----------------------------------------------------------------
Input size (MB): 0.00
Forward/backward pass size (MB): 0.65
Params size (MB): 1.61
Estimated Total Size (MB): 2.26
```

```
----------------------------------------------------------------
```

```
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")

model = CNNClassifier().to(device)

criterion = nn.CrossEntropyLoss()

optimizer = optim.Adam(model.parameters(), lr=0.001)
```

```
def train_model(model, train_loader, num_epochs=3):

    model.train()

    for epoch in range(num_epochs):

        running_loss = 0.0

        for images, labels in train_loader:

            images, labels = images.to(device), labels.to(device)

            optimizer.zero_grad()

            outputs = model(images)
            loss = criterion(outputs, labels)

            loss.backward()
            optimizer.step()

            running_loss += loss.item()

        print('Name: kaaviyan k')
        print('Register Number: 212224240066')
        print(f'Epoch [{epoch+1}/{num_epochs}], Loss: {running_loss/len(train_loader):.4f}')
```

```
train_model(model, train_loader)
```

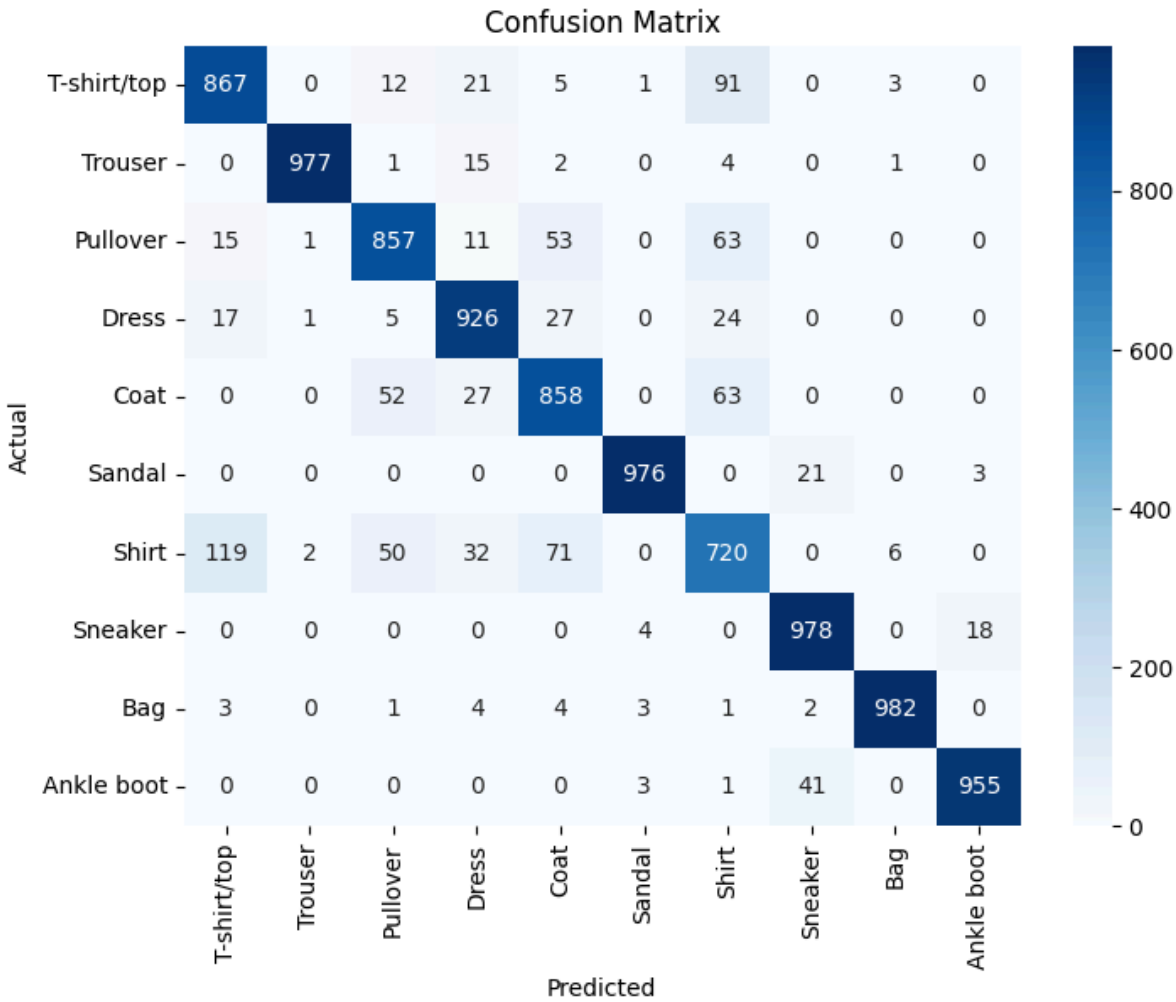
```
Name: kaaviyan k
Register Number: 212224240066
Epoch [1/3], Loss: 0.4479
Name: kaaviyan k
Register Number: 212224240066
Epoch [2/3], Loss: 0.2876
Name: kaaviyan k
Register Number: 212224240066
Epoch [3/3], Loss: 0.2429
```

```
def test_model(model, test_loader):

    model.eval()
    correct = 0
    total = 0
    all_preds = []
```

```
    all_labels = []

    with torch.no_grad():
        for images, labels in test_loader:

            images = images.to(device)
            labels = labels.to(device)

            outputs = model(images)
            _, predicted = torch.max(outputs, 1)

            total += labels.size(0)
            correct += (predicted == labels).sum().item()

            all_preds.extend(predicted.cpu().numpy())
            all_labels.extend(labels.cpu().numpy())

    accuracy = correct / total

    print('Name:kaaviyan k')
    print('Register Number: 212224240066')
    print(f'Test Accuracy: {accuracy:.4f}')

    cm = confusion_matrix(all_labels, all_preds)
    plt.figure(figsize=(8, 6))

    print('Name: kaaviyan k')
    print('Register Number: 212224240066')

    sns.heatmap(cm, annot=True, fmt='d', cmap='Blues',
                xticklabels=test_dataset.classes,
                yticklabels=test_dataset.classes)

    plt.xlabel('Predicted')
    plt.ylabel('Actual')
    plt.title('Confusion Matrix')
    plt.show()

    print('Name: kaaviyan k')
    print('Register Number: 212224240066')
    print("Classification Report:")
    print(classification_report(all_labels, all_preds,
                                target_names=test_dataset.classes))
```

```
test_model(model, test_loader)
```

```
Name:kaaviyan k
Register Number: 212224240066
Test Accuracy: 0.9096
Name: kaaviyan k
Register Number: 212224240066
```



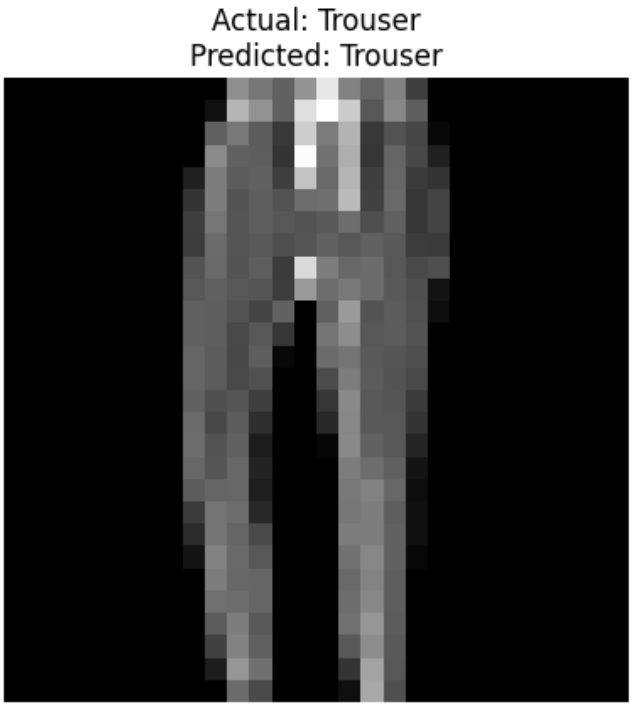

```
Name: kaaviyan k
Register Number: 212224240066
Classification Report:
              precision    recall  f1-score   support

 T-shirt/top       0.85      0.87      0.86      1000
     Trouser       1.00      0.98      0.99      1000
    Pullover       0.88      0.86      0.87      1000
       Dress       0.89      0.93      0.91      1000
        Coat       0.84      0.86      0.85      1000
      Sandal       0.99      0.98      0.98      1000
       Shirt       0.74      0.72      0.73      1000
     Sneaker       0.94      0.98      0.96      1000
         Bag       0.99      0.98      0.99      1000
  Ankle boot       0.98      0.95      0.97      1000

    accuracy                           0.91     10000
   macro avg       0.91      0.91      0.91     10000
weighted avg       0.91      0.91      0.91     10000
```

```
import matplotlib.pyplot as plt

def predict_image(model, image_index, dataset):
```

```
    model.eval()

    image, label = dataset[image_index]

    image = image.to(device)

    with torch.no_grad():
        output = model(image.unsqueeze(0))
        _, predicted = torch.max(output, 1)

    class_names = dataset.classes

    print('Name:kaaviyan k')
    print('Register Number: 212224240066')

    plt.imshow(image.cpu().squeeze(), cmap="gray")
    plt.title(f'Actual: {class_names[label]}\nPredicted: {class_names[predicted.item()]}')
    plt.axis("off")
    plt.show()

    print(f'Actual: {class_names[label]}, Predicted: {class_names[predicted.item()]}')
```

```
predict_image(model, image_index=80, dataset=test_dataset)
```

```
Name:kaaviyan k
Register Number: 212224240066
```




```
Actual: Trouser, Predicted: Trouser
```